

SOFTWARE ENGINEERING

EXAM ANSWERS

Easy English + Hinglish Explanations

20 Important Questions | Exam-Oriented Notes

Quick Revision: SDLC Models

Model	Core Idea
Classical Waterfall	Strictly sequential development.
Iterative Waterfall	Waterfall with limited feedback/backtracking.
Prototype	Build a sample to understand unclear requirements.
Evolutionary	Build and improve the real product through versions.
Spiral	Iterative development driven by risk analysis.

1. Classical Waterfall Model vs Prototype Model

Exam Answer

Classical Waterfall follows a strict sequence of phases. Requirements are mostly fixed at the beginning, customer involvement is limited after requirements, and working software is delivered late. Changes are difficult and costly. It is suitable when requirements are clear and stable.

Prototype Model first creates an early working model called a prototype. The customer evaluates it and gives feedback. Requirements can be refined before or during development. It is suitable when requirements are unclear.

Basis	Classical Waterfall	Prototype
Process	Strictly sequential	Prototype is created first
Requirements	Mostly fixed early	Discovered/refined through feedback
Customer involvement	Lower after requirements	High
Delivery	Working software later	Early prototype
Changes	Difficult and costly	Easier during requirement discovery

Hinglish Explanation

Waterfall ko seedhi staircase samjho: Requirement -> Design -> Coding -> Testing. Ek phase ke baad next phase.

Prototype me pehle sample banate ho, jaise restaurant ki sample dish. Customer dekhkar bolta hai kya change chahiye.

2. Core Objectives of Software Engineering and Need for a Systematic Approach

Exam Answer

The main objectives of Software Engineering are: (1) develop high-quality software, (2) satisfy user requirements, (3) complete the project within time and budget, (4) make software reliable, maintainable and scalable, and (5) reduce errors, risks and development cost.

A systematic approach is necessary for large-scale software because such projects have many people, modules, requirements and dependencies. Without proper planning and processes, projects can suffer from defects, delays, cost overruns and failure.

Hinglish Explanation

Simple goal: right software + right quality + right time + right cost.

Chhote project me informal work chal sakta hai, lekin Amazon ya hospital jaisa project bina process ke control se bahar ho jayega. Software banana ghar banane jaisa hai: map aur plan chahiye.

3. Software Development Life Cycle (SDLC) and Deliverables

Exam Answer

SDLC is a systematic process used to develop, deliver and maintain software from the initial idea to maintenance.

Typical phases and deliverables are:

1. Planning/Feasibility - Feasibility Report or Project Plan.
2. Requirement Analysis - SRS.
3. System Design - Design Document.
4. Implementation - Source Code/Build.
5. Testing - Test Cases and Test Reports.
6. Deployment - Released/Installed Software.
7. Maintenance - Updated Versions and Maintenance Records.

Hinglish Explanation

SDLC ko software ki life journey samjho: Idea -> Requirements -> Design -> Coding -> Testing -> Deployment -> Maintenance.

Deliverable ka matlab phase ke end me milne wala official output/document.

4. Classical Waterfall Model vs Iterative Waterfall Model

Exam Answer

Classical Waterfall is strictly sequential and assumes that a phase is completed before the next phase starts. It provides very little support for going back to an earlier phase.

Iterative Waterfall keeps the waterfall sequence but allows feedback and limited backtracking to earlier phases. This makes it more flexible and allows defects or requirement issues to be corrected before final delivery.

Basis	Classical Waterfall	Iterative Waterfall
Flow	Strictly sequential	Sequential with feedback
Backtracking	Very limited	Allowed to previous phases
Defect handling	Late fixes are costly	Earlier-phase correction possible
Flexibility	Low	Higher

Hinglish Explanation

Classical: A -> B -> C -> D.

Iterative: A -> B -> C -> D, aur zarurat pade to D se C ya C se B wapas ja sakte ho. Isliye iterative version real projects ke liye zyada practical hai.

5. Importance of Risk Analysis in Spiral Model

Exam Answer

The Risk Analysis quadrant is the most important part of the Spiral Model. It identifies possible risks, estimates their probability and impact, studies alternatives, and plans actions to reduce or remove the risks. Development continues with a better understanding of uncertainty.

Hinglish Explanation

Spiral ka simple rule hai: pehle poochho risk kya hai, phir build karo. Example: hospital system me patient-data leak ka risk. Encryption, access control and backup jaise solutions risk ko reduce karte hain.

6. Legal and Operational Feasibility

Exam Answer

Legal Feasibility checks whether the proposed system follows applicable laws, regulations, contracts and legal obligations. Example: a healthcare system must comply with applicable patient-data privacy and security requirements.

Operational Feasibility checks whether the organization can use and operate the proposed system effectively. Example: a hospital may reject a technically excellent system if doctors and nurses find it too difficult to use.

Hinglish Explanation

Legal = kya system legally allowed hai?

Operational = kya log is system ko daily work me practically use kar paayenge?

Car ka road par legal hona legal feasibility hai; driver ka usse safely chala paana operational side hai.

7. Cost-Benefit Analysis for Hospital Management System

Exam Answer

For economic feasibility, identify and estimate all major costs and benefits, convert them to monetary values where reasonable, and compare the total cost with expected benefits over a chosen period.

Costs may include development, servers/hardware, training, security and maintenance. Benefits may include reduced paperwork, fewer billing errors, staff time savings, faster registration and better resource use.

Example: Total implementation cost = Rs. 20 lakh. Estimated annual benefit = Rs. 16 lakh. The organization can compare the investment with the expected yearly savings/benefits and estimate the payback period.

Hinglish Explanation

Simple question: kitna paisa lagega vs kitna benefit milega?

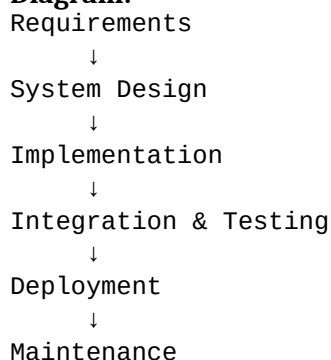
Jaise 1 lakh ka machine kharidkar har saal 50,000 save ho rahe hain. Yehi basic cost-benefit thinking hai.

8. Architecture of Classical Waterfall Model and Its Limitations

Exam Answer

The Classical Waterfall Model follows a linear, sequential structure.

Diagram:



Why unrealistic for modern dynamic projects:

- Requirements can change after development starts.
- Late defects and changes can be expensive.
- Customer feedback is limited during development.
- Working software is often delivered late.

Hinglish Explanation

Waterfall ko paani ki waterfall ki tarah yaad rakho: requirements se maintenance tak process mostly one-way flow karta hai.

Problem: modern projects me requirements change hoti rehti hain. Agar late stage par client new feature maange, to pehle wale phases ko change karna difficult aur expensive ho sakta hai.

9. Feedback Loops in Iterative Waterfall and Defect Handling

Exam Answer

The Iterative Waterfall Model permits feedback from a later phase to an earlier phase. If testing reveals a defect caused by a wrong design or requirement, the team can return to that earlier phase, correct the problem and continue development.

In the Classical Waterfall Model, such backtracking is not formally supported and late defects are more costly to fix.

Hinglish Explanation

Testing me pata chala database design hi wrong hai. Iterative model me Testing -> Design -> Correction -> Coding -> Testing ho sakta hai.

Classical model me reverse movement major difficulty hoti hai.

10. When to Choose an Iterative Model

Exam Answer

Choose an iterative model when requirements are likely to change, customer feedback is important, the project has uncertainty or complexity, early working versions are useful, or risks must be handled gradually.

Choose a strictly sequential model when requirements are stable, well understood and unlikely to change.

Hinglish Explanation

Changing requirements -> Iterative.

Stable requirements -> Waterfall.

Startup app generally gets frequent feedback, while a well-defined internal payroll system may have more stable requirements.

11. Verification vs Validation

Exam Answer

Verification checks whether the software is being built according to specified requirements, standards and design. It often uses reviews and inspections.

Validation checks whether the final software actually satisfies user needs and intended use. It usually involves executing the software.

Verification example without code execution: reviewing the SRS or design document.

Validation example involving execution: running a login test on the application.

Hinglish Explanation

Verification = "Are we building the product right?"

Validation = "Are we building the right product?"

Yaad rakho: verification specification ke against check hota hai; validation user ki actual need ke against.

12. Dating App with Immediate User Feedback - Suitable Model

Exam Answer

An Evolutionary/Iterative approach is suitable because the app features may change quickly based on real user feedback. The team can release a small working version, collect feedback, and improve the product in the next version.

Example: V1 - profile and login; V2 - matching; V3 - chat; V4 - improved matching based on feedback.

Hinglish Explanation

Build -> Release -> Feedback -> Improve -> Repeat.

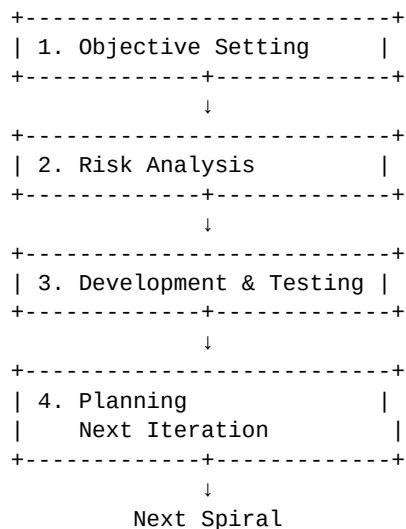
Yahan "perfect first version" banane ki koshish inefficient hai because startup ko initially hi sab user needs ka exact idea nahi hota.

13. Spiral Model and Its Four Quadrants

Exam Answer

The Spiral Model develops software through repeated cycles. Each cycle has four major activities: (1) Objective Setting - define objectives, alternatives and constraints; (2) Risk Analysis - identify and reduce risks; (3) Development and Testing - build and test the product; and (4) Planning - plan the next iteration.

Exam-friendly diagram:



Hinglish Explanation

Har spiral round me: Goal -> Risk -> Build/Test -> Next Plan. Phir next round. Isliye spiral gradually larger and better hota jata hai.

14. Prototype Model vs Evolutionary Model

Exam Answer

Prototype Model mainly creates a sample to discover or clarify requirements. The prototype may be discarded or rebuilt. User involvement is usually high.

Evolutionary Model develops the actual product in successive working versions. User feedback is used continuously, and the structure evolves rather than being thrown away. Costs are spread across iterations.

Basis	Prototype	Evolutionary
User involvement	High	High/continuous
Cost	Initial prototype cost; may be discarded	Cost spread over versions

Basis	Prototype	Evolutionary
Structure	Prototype may be rebuilt	Actual product evolves
Main purpose	Clarify requirements	Gradually build final product

Hinglish Explanation

Prototype = cardboard model of a house; final house alag ban sakta hai.

Evolutionary = first floor banao, phir second floor, phir third floor; same product grow karta hai.

15. Why Spiral Model is a Meta-Model

Exam Answer

The Spiral Model is called a meta-model because it can incorporate characteristics of other lifecycle approaches within different cycles. A spiral iteration may use waterfall-style activities, prototyping, incremental development or evolutionary development. Its defining focus is continuous risk analysis.

Hinglish Explanation

Meta-model ko “model ke upar model” samjho. Spiral ek fixed recipe force nahi karta; project ki need ke hisaab se different techniques combine kar sakta hai.

16. Characteristics of a Good SRS

Exam Answer

A good SRS should be correct, complete, unambiguous, consistent, verifiable/testable, modifiable and traceable.

A requirement should be precise. For example, “the system should be fast” is weak; “search results should appear within 2 seconds” is measurable and testable.

Hinglish Explanation

SRS software ka official instruction document hai. Ambiguous statement problem create karta hai. “Fast system” subjective hai; “within 2 seconds” clear hai.

17. Functional vs Non-Functional Requirements

Exam Answer

Functional Requirements describe what the system should do. Examples for an e-commerce platform: (1) user can create an account, (2) user can add products to the cart, (3) user can place an order.

Non-Functional Requirements describe how well or under what quality constraints the system should work.

Examples: (1) page should load within 2 seconds, (2) system should provide 99.9% availability, (3) customer data should be encrypted.

Basis	Functional	Non-Functional
Question	What should the system do?	How well should it work?
Examples	Login, search, order	Speed, security, availability

Hinglish Explanation

Functional = WHAT the system does.

Non-functional = HOW WELL it does it.

Login, search, add-to-cart are functions. Fast, secure, reliable are quality requirements.

18. Dating App with Immediate User Feedback - Suitable Model

Exam Answer

The Evolutionary/Iterative Model is appropriate because features will evolve from immediate user feedback. Working versions can be released quickly and improved continuously.

Hinglish Explanation

Same logic as Q12: Build -> Feedback -> Improve. A prototype alone is not enough because the actual product needs continuous evolution.

19. SRS as a Legal Contract and Consequences of Poor SRS

Exam Answer

The SRS acts as a common agreement between the client and development team because it defines the scope, required behavior and expected deliverables. It can be used as a reference during development, testing, acceptance and dispute resolution.

A poorly written SRS can cause misunderstanding, scope creep, extra cost, delays, more defects, customer dissatisfaction and legal disputes.

Hinglish Explanation

SRS basically says: client kya maang raha hai aur team kya deliver karegi. Agar "payment system" likha hai but UPI, card, wallet, EMI kuch define nahi, later conflict hoga.

20. Evolutionary Model of Software Development

Exam Answer

The Evolutionary Model develops software gradually through multiple working versions. A basic version is delivered first, user feedback is collected, and new features and improvements are added in later versions.

Benefits include early delivery, continuous feedback, easier handling of changing requirements and reduced risk.

Example: V1 - login and core features; V2 - search and categories; V3 - recommendations; V4 - more advanced features.

Hinglish Explanation

Evolutionary model ko growth jaisa samjho: basic product -> improved product -> mature product. Har version usable hota hai, aur next version previous version par build karta hai.

One-Line Definitions for Last-Minute Revision

SDLC: A systematic process used to develop, deliver and maintain software.

Waterfall Model: A sequential model in which each phase is completed before the next phase begins.

Iterative Waterfall: Waterfall with feedback and limited backtracking to earlier phases.

Prototype Model: A model that creates an early sample to understand and refine requirements.

Evolutionary Model: A model that develops and improves the real software through successive working versions.

Spiral Model: A risk-driven iterative model that emphasizes repeated risk analysis.

SRS: A document specifying the functional and non-functional requirements of a software system.

Verification: Checking whether the product is being built according to specifications.

Validation: Checking whether the developed product satisfies user needs.

Functional Requirement: A requirement describing what the system should do.

Non-Functional Requirement: A requirement describing quality or performance constraints.

Feasibility Study: An analysis to determine whether a proposed project is practical and worthwhile.